SOFTWARE REQUIREMENTS SPECIFICATION

RSwitch Data Intelligence Platform

(RDIP)

Project Name	RSwitch Data Intelligence Platform (RDIP)
Document Version	v1.1
Date	2026-07-24
Status	Draft — Updated per Stakeholder Feedback
Authors	Project Team
Stakeholders	RSwitch Board, Compliance Division, Operations Team, Participating Financial Institutions

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1. Document Information

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Revision History

Version	Date	Author	Description
v0.1	2026-07-10	Project Team	Initial draft
v1.0	2026-07-22	Project Team	Updated based on RSwitch feedback — reframed from customer to participant model; removed participant portal
v1.1	2026-07-24	Project Team	Updated per Raica Kwizera feedback — clarified S&M scope, email integration, AI baseline, risk categories, data flexibility, dynamic filtering, resubmission workflow, alert lifecycle rules (skip-prevention, auto-escalation, closure requirements)

2. Introduction

2.1 Purpose

This document defines all functional and non-functional requirements of the RSwitch Data Intelligence Platform (RDIP) so that designers, developers, testers, and stakeholders have one authoritative source of truth. It captures requirements derived from stakeholder feedback and governs the design, development, and acceptance of the system.

2.2 Scope

RDIP is an internal operational platform designed for RSwitch and its participating financial institutions. It consolidates data integration monitoring, participant compliance tracking, governance coordination, AI-assisted operational insights, and financial inclusion reporting into a single unified system.

The platform serves RSwitch staff across compliance, business, and technical functions. It does not serve individual end-customers of financial institutions directly.

2.3 Intended Audience

Audience	Purpose
RSwitch Board & Management	Review governance modules and strategic reports
Compliance Division	Monitor participant compliance and deadlines
Business / Operations Team	Track integration status, financial inclusion data
Technical / IT Team	Manage integrations, pipelines, and data mappings
Developers	Implement features from this specification
QA / Testers	Validate system against these requirements
Participating Institutions	Submit data, view their own compliance status

2.4 Business Objectives

- Give RSwitch a single operational view of all participating financial institutions.
- Replace manual, fragmented data collection with structured submission workflows that can progressively automate over time.
- Enable proactive compliance monitoring with automated deadline tracking and escalation.
- Provide a central governance repository for the board, committees, and working groups.
- Support AI-assisted identification of unusual patterns and operational risks across participants.
- Produce financial inclusion reports segmented by institution type and region.

2.5 Project Vision

RSwitch operates as Rwanda's national payment switch, connecting commercial banks, e-money issuers, microfinance institutions, and SACCOs. Currently, operational data about participants is scattered across spreadsheets, emails, and disconnected systems. RDIP becomes the authoritative hub where RSwitch staff can monitor every participant's integration health, compliance standing, and operational risk — with a clear path from

manual data submissions today to fully automated data exchange in the future.

2.6 Assumptions

- Participating institutions will initially submit data manually (uploads/forms); automation is introduced progressively.
- RSwitch will provide sample data for the development team to generate realistic synthetic datasets.
- The platform is a web application accessible via modern browsers; no mobile app is in scope for v1.
- RSwitch controls all user accounts; participant institutions do not self-register.
- AI features will use historical operational data provided by RSwitch; no specific external ML model has been mandated.

2.7 Constraints

- The platform must comply with Rwanda's data protection regulations and any applicable financial sector regulatory requirements.
- The system must not store or display individual customer-level transaction data unless specifically approved by RSwitch.
- Initial deployment is on RSwitch's internal infrastructure; cloud migration is out of scope for v1.
- Sample data for development must be provided by the RSwitch team before dashboard finalization.

2.8 Out of Scope

- Public-facing portals for individual banking customers.
- Core payment switching or transaction processing (RDIP monitors and reports; it does not process payments).
- Mobile applications (iOS / Android).
- Direct integration with external international payment networks in v1.
- Automated credit scoring or lending decisions for individual customers.

3. System Overview

3.1 What the System Is

RDIP is a web-based operational intelligence and governance platform for RSwitch. It centralizes information from all participating financial institutions — their integration health, data submissions, compliance status, governance documents, and operational metrics — into one secure, role-based interface.

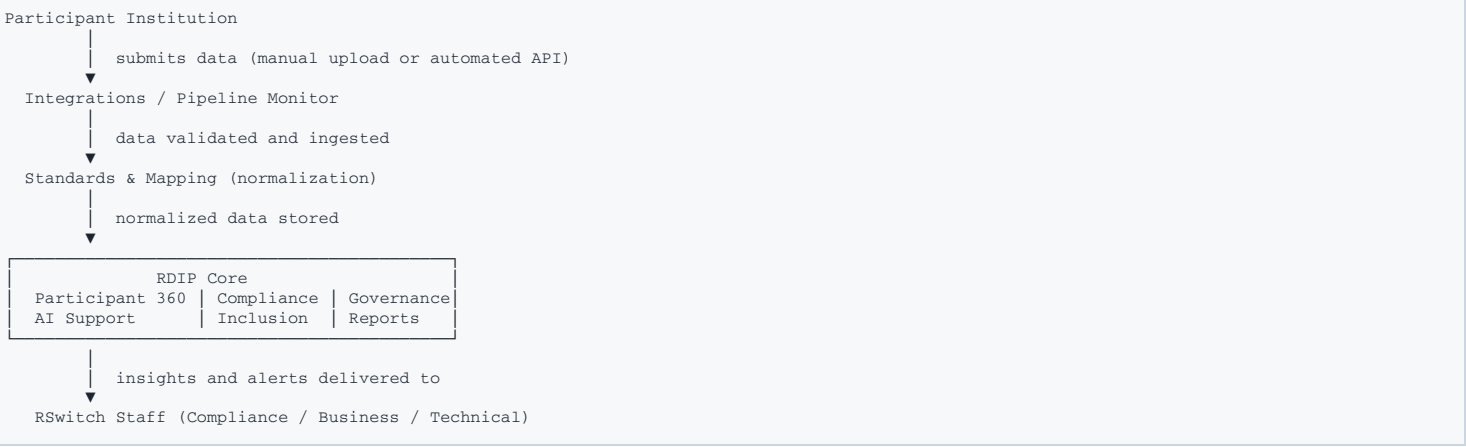
3.2 What Problem It Solves

Problem Today	How RDIP Solves It
Participant data spread across emails/spreadsheets	Central submission and monitoring hub
No visibility into integration failures in real time	Pipeline Monitor with live status per institution
Compliance deadlines tracked manually	Automated deadline tracking, reminders, and escalation
Governance documents stored in scattered folders	Central document repository with calendar and action items
No structured way to identify unusual participant behavior	AI-assisted pattern detection and prioritization
Financial inclusion metrics require manual aggregation	Automated reporting by institution type and region

3.3 Who Uses It

- **RSwitch Compliance Team** — Monitor regulatory deadlines, incidents, participant compliance history.
- **RSwitch Business / Operations Team** — Track financial inclusion metrics, review AI insights, generate reports.
- **RSwitch Technical / Admin Team** — Configure integrations, manage pipelines, define data mappings, manage users.

3.4 High-Level Workflow



4. User Types / Roles

4.1 Role Definitions

RDIP uses three primary user groups as agreed with stakeholders. Access is dynamic and permission-based.

Role	Label	Description
compliance	Compliance User	Monitors participant compliance, deadlines, incidents, and remediation actions
business	Business User	Reviews reports, financial inclusion analytics, governance records, and AI insights
technical	Technical / Admin User	Manages integrations, pipelines, data mappings, system settings, and user accounts

4.2 Permission Overview

Module	Compliance	Business	Technical
Dashboard	View	View	View
Integrations	View	View	Manage
Pipeline Monitor	View	View	Manage
Standards & Mapping	View	—	Manage
Participant 360	View	View	View
Governance	Manage	View	View
Compliance Monitoring	Manage	View	View
AI Decision Support	View	Manage	View
Financial Inclusion	View	Manage	View
Reports	Manage	Manage	View
Notifications	View	View	Manage
Users & Security	—	—	Manage
Settings	—	View	Manage
Audit Log	View	—	Manage

Manage = read + write + create + delete. View = read-only.

5. Functional Requirements

5.1 Authentication & Access Control (AUTH)

ID	Requirement
REQ-AUTH-001	The system SHALL allow users to log in using an email address and password.
REQ-AUTH-002	The system SHALL enforce Multi-Factor Authentication (MFA) via a 6-digit TOTP code for all users.
REQ-AUTH-003	The system SHALL lock an account after 5 consecutive failed login attempts for 30 minutes.
REQ-AUTH-004	The system SHALL allow users to reset their password via a time-limited link sent to their registered email (link expires in 60 minutes).
REQ-AUTH-005	The system SHALL maintain a session timeout of 30 minutes of inactivity, after which the user must re-authenticate.
REQ-AUTH-006	The system SHALL support a "Trust this device for 30 days" option on MFA step.
REQ-AUTH-007	The system SHALL log every login attempt (success and failure) with timestamp, IP address, and user agent.
REQ-AUTH-008	The system SHALL enforce role-based access control (RBAC) so that each user can only access modules permitted by their role.
REQ-AUTH-009	The system SHALL allow Technical/Admin users to assign and modify user roles.
REQ-AUTH-010	The system SHALL display an access-denied message when a user attempts to access a module outside their permissions, without revealing that the module exists.

5.2 Dashboard (DASH)

ID	Requirement
REQ-DASH-001	The system SHALL display a role-aware dashboard as the default landing page after login.
REQ-DASH-002	The dashboard SHALL display KPI widgets relevant to the logged-in user's role (see Section 15).
REQ-DASH-003	The dashboard SHALL display an Attention Queue listing the top unresolved alerts across compliance, integrations, and incidents.
REQ-DASH-004	The dashboard SHALL display a summary of integration health across all participants (healthy / warning / failed count).
REQ-DASH-005	The dashboard SHALL display upcoming compliance deadlines within the next 30 days.
REQ-DASH-006	The dashboard SHALL refresh KPI data automatically every 5 minutes when auto-refresh is enabled.
REQ-DASH-007	The dashboard SHALL allow users to dismiss individual attention queue items for their current session.
REQ-DASH-008	The dashboard SHALL display a Recent Activity log showing the last 10 system events relevant to the user's role.
REQ-DASH-009	Dashboard widgets SHALL support dynamic filtering by: institution name, participant type (Commercial Bank / E-money Issuer / MFI / SACCO), region, status, date range, and risk category. Filters SHALL persist within a user session.
REQ-DASH-010	The dashboard SHALL NOT rely solely on fixed, pre-built views. Users SHALL be able to adjust the period and scope of each widget without navigating away from the dashboard.

5.3 Integrations (INTG)

ID	Requirement
REQ-INTG-001	The system SHALL maintain a catalog of all participating financial institutions and their configured data sources.
REQ-INTG-002	The system SHALL support two integration modes: (a) Manual Upload — institution submits files via the platform; (b) Automated API — system pulls data from institution's endpoint on a schedule.
REQ-INTG-003	The system SHALL display the current status of each integration: Healthy, Warning, Failed, Pending, or Not Configured.
REQ-INTG-004	The system SHALL display the last successful data transfer date/time for each integration.
REQ-INTG-005	The system SHALL allow Technical/Admin users to add new participant integrations, specifying: institution name, integration type, endpoint URL (for API mode), authentication credentials (stored encrypted), and expected submission schedule.
REQ-INTG-006	The system SHALL allow Technical/Admin users to trigger a manual sync for any configured automated integration.
REQ-INTG-007	The system SHALL allow Technical/Admin and Compliance users to upload data files (CSV, Excel, XML) on behalf of a participant institution when in Manual Upload mode.
REQ-INTG-008	The system SHALL validate uploaded files against the expected schema before accepting them, and return a structured error report if validation fails.
REQ-INTG-009	The system SHALL display, for each integration, the volume of records transferred in the last submission and the number of validation errors.
REQ-INTG-010	The system SHALL send an automated notification to the Technical/Admin team when an integration fails or a submission is overdue.
REQ-INTG-011	The system SHALL maintain a submission history log per participant showing all past transfers, their status, record counts, and error counts.

5.4 Pipeline Monitor (PIPE)

ID	Requirement

REQ-PIPE-001	The system SHALL display all active and recent data ingestion pipeline jobs in a table view.
REQ-PIPE-002	The system SHALL show for each pipeline job: job name, source institution, status, progress percentage, start time, duration, records processed, and error count.
REQ-PIPE-003	Pipeline job statuses SHALL be: Running, Completed, Failed, Queued, Paused.
REQ-PIPE-004	The system SHALL allow users to expand a pipeline job to view its step-by-step execution log.
REQ-PIPE-005	The system SHALL allow Technical/Admin users to trigger a Reprocess action on failed pipeline jobs.
REQ-PIPE-006	The system SHALL allow Technical/Admin users to stop a currently running pipeline job.
REQ-PIPE-007	The system SHALL allow filtering of pipeline jobs by status and by source institution.
REQ-PIPE-008	The system SHALL allow Technical/Admin users to create new pipeline configurations specifying: source, schedule (cron expression), and target mapping.
REQ-PIPE-009	The system SHALL retain pipeline execution logs for a minimum of 90 days.
REQ-PIPE-010	The system SHALL NOT display individual customer-level transaction records in the pipeline monitor; only aggregate counts and metadata SHALL be shown.

5.5 Standards & Mapping (MAP)

> ***Scope clarification (per stakeholder feedback):*** *Standards & Mapping is primarily a backend / data integration function. It does not need to be a prominent standalone module for business or compliance users. It is exposed as a Technical/Admin-only configuration interface — accessible under Settings or as a supporting tool within Integrations — allowing the technical team to define and version how incoming data is normalized. Business and Compliance users have no need to interact with this module directly.*

ID	Requirement
REQ-MAP-001	The system SHALL allow Technical/Admin users to create field-level mapping definitions that translate source data fields to a canonical target format (e.g. ISO 20022, JSON Canonical).
REQ-MAP-002	Each mapping SHALL have: a unique mapping ID, name, source system, target standard, version number, status, and a count of mapped vs total fields.
REQ-MAP-003	Mapping statuses SHALL be: Healthy (all fields mapped), Warning (some fields unmapped), Draft (not yet published).
REQ-MAP-004	The system SHALL allow Technical/Admin users to publish a mapping as a versioned release (e.g. v1.0, v2.0).
REQ-MAP-005	The system SHALL maintain version history for each mapping, showing who published each version and when.
REQ-MAP-006	The system SHALL allow Technical/Admin users to clone an existing mapping as the starting point for a new one.
REQ-MAP-007	The system SHALL apply the active published mapping version during pipeline ingestion automatically.
REQ-MAP-008	The system SHALL display a warning when a mapping has unmapped fields before allowing it to be published.
REQ-MAP-009	The Standards & Mapping interface SHALL be accessible only to Technical/Admin users and SHALL be positioned as a configuration tool within the Integrations or Settings area, not as a primary navigation module.

5.6 Participant 360 (P360)

ID	Requirement
REQ-P360-001	The system SHALL provide a unified participant profile page for each registered financial institution.
REQ-P360-002	The participant profile SHALL display: institution name, type (commercial bank / e-money issuer / MFI / SACCO), registration date, assigned contact, and current overall status.
REQ-P360-003	The participant profile SHALL display Integration Status: current health of each connected data source for that institution.
REQ-P360-004	The participant profile SHALL display Data Submission History: a chronological log of all data submissions, their status, record counts, and error counts.
REQ-P360-005	The participant profile SHALL display Compliance Status: a summary of required compliance documents, their submission status, and upcoming or overdue deadlines.
REQ-P360-006	The participant profile SHALL display Incidents: a list of all open and resolved incidents associated with this institution.
REQ-P360-007	The participant profile SHALL display Technical Performance: failure rates, uptime percentage, and average data submission latency over the last 30 days.
REQ-P360-008	The participant profile SHALL display Outstanding Actions: a list of pending items assigned to either RSwitch or the institution.
REQ-P360-009	The participant profile SHALL display a Participant Risk Assessment with clearly defined inputs and scoring methodology visible to authorized users.
REQ-P360-010	The Participant Risk Assessment SHALL be broken into three distinct risk categories: (a) Compliance Risk — submission adherence rate, overdue deadlines, rejection history; (b) Operational Risk — incident frequency, severity, and resolution time; (c) Technical Risk — integration uptime, pipeline failure rate, average submission latency. An overall composite score SHALL also be displayed for quick reference.
REQ-P360-011	The risk scoring weights, thresholds, and indicators SHALL be configurable by Technical/Admin users through the Settings module, allowing the methodology to evolve as data and business understanding matures. The SRS does not mandate fixed weights; these SHALL be determined during the design phase.
REQ-P360-012	A risk score SHALL only be displayed when a minimum of 3 months of submission history is available for the participant. Where insufficient

	history exists, the system SHALL display "Insufficient data — score pending."
REQ-P360-013	The system SHALL NOT display individual customer-level indicators (e.g. probability of default per customer) in the Participant 360 view.
REQ-P360-014	The system SHALL allow Compliance users to add notes or annotations to a participant's profile.

5.7 Governance (GOV)

ID	Requirement
REQ-GOV-001	The system SHALL provide a central Governance module serving the Board, Steering Committees, and Technical/Operational Working Groups.
REQ-GOV-002	The system SHALL allow authorized users to upload governance documents (PDF, Word, Excel) and tag them by category (policy, minutes, decision, working paper).
REQ-GOV-003	Uploaded documents SHALL have: title, category, upload date, uploader, version, associated committee/group, and access level.
REQ-GOV-004	The system SHALL maintain a Meeting Calendar displaying scheduled meetings for all committees and working groups.
REQ-GOV-005	The system SHALL allow authorized users to create meeting records containing: date, attendees, agenda, minutes document, and a list of decisions and action items.
REQ-GOV-006	Each action item SHALL have: description, assigned owner (RSwitch user or institution), due date, and status (Open / In Progress / Completed / Overdue).
REQ-GOV-007	The system SHALL display a dashboard of overdue action items grouped by committee or working group.
REQ-GOV-008	The system SHALL maintain a register of working-group membership, showing which RSwitch staff and participant representatives belong to each group.
REQ-GOV-009	The system SHALL send automated reminders to action item owners 7 days and 1 day before the due date.
REQ-GOV-010	The system SHALL send an escalation notification to the committee chair when an action item becomes overdue.
REQ-GOV-011	Governance module access SHALL be restricted: Business users view only; Compliance users manage; Technical users view only.

5.8 Compliance Monitoring (COMP)

ID	Requirement
REQ-COMP-001	The system SHALL maintain a Compliance Register listing all required compliance documents per participant institution.
REQ-COMP-002	Each compliance requirement entry SHALL have: requirement name, regulatory reference, frequency (monthly / quarterly / annual), responsible participant, due date, and submission status.
REQ-COMP-003	Compliance submission statuses SHALL be: Pending, Submitted, Under Review, Approved, Rejected, Overdue.
REQ-COMP-004	The system SHALL display a compliance calendar showing all upcoming deadlines within the next 90 days.
REQ-COMP-005	The system SHALL highlight overdue compliance submissions in red and send automated reminders to the responsible participant.
REQ-COMP-006	The system SHALL allow Compliance users to record incidents per participant, including: incident date, description, severity (Low / Medium / High / Critical), and assigned remediation owner.
REQ-COMP-007	Incident statuses SHALL be: Open, Under Investigation, Remediation In Progress, Resolved, Closed.
REQ-COMP-008	The system SHALL track failure rates per participant: percentage of submissions that were rejected or contained errors over a rolling 12-month period.
REQ-COMP-009	The system SHALL maintain a full compliance history per participant, showing all past submissions, their outcomes, and any associated incidents.
REQ-COMP-010	The system SHALL allow Compliance users to define remediation action items for unresolved incidents, including owner and due date.
REQ-COMP-011	The system SHALL send escalation notifications to senior Compliance users when an incident remains unresolved past its remediation due date.

5.9 AI Decision Support (AI)

ID	Requirement
REQ-AI-001	The system SHALL provide an AI-assisted Decision Support module that helps RSwitch staff identify unusual operational patterns across participant institutions.
REQ-AI-002	AI outputs SHALL be advisory and explainable; the system SHALL NOT make automated final decisions or automated enforcement actions.
REQ-AI-003	The system SHALL display AI-generated alerts identifying participants exhibiting unusual behavior (e.g. sudden spike in submission errors, unexplained failure rate change).
REQ-AI-004	Each AI alert SHALL include: the participant name, the indicator that triggered the alert, the baseline value used, the current observed value, the deviation, and a recommended area for investigation.
REQ-AI-005	The system SHALL provide an AI-generated priority list of participants requiring attention, ranked by composite risk signal strength.
REQ-AI-006	The system SHALL allow Business and Compliance users to manage AI alerts through a defined lifecycle: Generated → Acknowledged → Under Investigation → Resolved / Escalated → Closed. Each status transition SHALL require a mandatory note from the acting user.
REQ-AI-006a	The system SHALL enforce that an alert CANNOT be moved directly from Generated to Under Investigation. It MUST pass through Acknowledged first, ensuring no alert is silently acted upon without a logged awareness note.

REQ-AI-006b	The system SHALL automatically escalate an alert to Escalated status and notify a senior Compliance or Business user if the alert remains at Acknowledged status without progressing for a configurable number of days (default: 3 days). Technical/Admin users SHALL be able to adjust this escalation threshold in Settings.
REQ-AI-006c	The system SHALL require a written resolution summary (minimum 20 characters) before an alert can be moved to Closed status. A closed alert SHALL be read-only and cannot be reopened. If the same issue recurs, a new alert SHALL be generated.
REQ-AI-007	The system SHALL display compliance and operational trend summaries, covering the last 3 months across all participants.
REQ-AI-008	All AI outputs SHALL be linked to the underlying data indicators so users can drill down to the source data.
REQ-AI-009	The system SHALL document and display the data sources and baseline methodology used to produce each AI output.
REQ-AI-010	The platform architecture SHALL be model-agnostic: the analytical approach (threshold-based, statistical, or ML-based) SHALL NOT be hardcoded and SHALL be replaceable without requiring application code changes. The specific model or approach SHALL be selected during the design phase based on available historical data and finalized business requirements.
REQ-AI-011	Baseline establishment SHALL follow a phased approach: (a) Release 1 — configurable predefined thresholds set by Technical/Admin users; (b) Release 2 — rolling historical baselines calculated from accumulated submission data (minimum 6 months); (c) Release 3 — peer-group comparison baselines across institutions of the same type. The system SHALL display which baseline method is active for each alert.
REQ-AI-012	Technical/Admin users SHALL be able to configure alert sensitivity thresholds (e.g. deviation percentage that triggers an alert) through the Settings module without requiring a code change.

5.10 Financial Inclusion (FIN)

ID	Requirement
REQ-FIN-001	The system SHALL provide financial inclusion reporting dashboards based on data submitted by participant institutions.
REQ-FIN-002	Financial inclusion reports SHALL support segmentation by institution type: E-money Issuers, Commercial Banks, Microfinance Institutions (MFIs), SACCOs.
REQ-FIN-003	Financial inclusion reports SHALL support segmentation by region where demographic data is available.
REQ-FIN-004	The system SHALL display aggregated metrics per participant category including, where data permits: number of accounts, active users, transaction volumes, and access rates.
REQ-FIN-005	The system SHALL allow Business users to filter reports by institution type, region, and time period.
REQ-FIN-006	The system SHALL clearly label each metric with its data source and the date it was last updated.
REQ-FIN-007	Dashboard metrics and visualizations SHALL be treated as indicative at this stage. Final data fields, metrics, and visualizations SHALL be confirmed once RSwitch has validated data availability with participating institutions. The module structure SHALL be designed to accommodate new metrics without requiring architectural changes.
REQ-FIN-008	The system SHALL allow Business users to export financial inclusion reports in PDF and Excel formats.

5.11 Reports (RPT)

ID	Requirement
REQ-RPT-001	The system SHALL allow authorized users to generate reports across the following categories: Compliance Summary, Integration Health, Participant Performance, Financial Inclusion, Incident Report, Governance Action Items.
REQ-RPT-002	Each report type SHALL have configurable filters (see Section 14).
REQ-RPT-003	The system SHALL allow reports to be exported in PDF and Excel (XLSX) formats.
REQ-RPT-004	The system SHALL maintain a report history showing all generated reports with the name, type, date, generated by, and download link.
REQ-RPT-005	The system SHALL allow Compliance and Business users to schedule recurring reports (daily / weekly / monthly) with automatic email delivery.
REQ-RPT-006	All reports SHALL support dynamic filtering by: institution, participant type, region, status, date range, and risk level. Filter combinations SHALL be saveable as named report templates.

5.12 Notifications (NOTIF)

ID	Requirement
REQ-NOTIF-001	The system SHALL support in-app notifications displayed in a notification bell in the top navigation bar.
REQ-NOTIF-002	The system SHALL support email notifications for time-sensitive events as defined in Section 13.
REQ-NOTIF-003	The system SHALL allow Technical/Admin users to configure which notification events are active.
REQ-NOTIF-004	The system SHALL display a notification history for the last 90 days.
REQ-NOTIF-005	The system SHALL allow users to mark individual notifications as read or mark all as read.

5.13 Users & Security (USR)

ID	Requirement
REQ-USR-001	The system SHALL allow Technical/Admin users to invite new users by email, specifying their role.

REQ-USR-002	The system SHALL send an invitation email with a time-limited registration link (expires in 48 hours).
REQ-USR-003	The system SHALL allow Technical/Admin users to deactivate user accounts without deleting them, preserving audit history.
REQ-USR-004	The system SHALL allow Technical/Admin users to change a user's role.
REQ-USR-005	The system SHALL display a user management table showing: name, email, role, status (active/inactive), and last login date.
REQ-USR-006	The system SHALL maintain an audit log of all user management actions (invite, role change, deactivation).

5.14 Settings (SET)

ID	Requirement
REQ-SET-001	The system SHALL allow Technical/Admin users to configure organization-level settings: organization name, logo, default timezone, and default date format.
REQ-SET-002	The system SHALL allow Technical/Admin users to manage API keys for external integrations (create, view masked, revoke).
REQ-SET-003	The system SHALL allow Technical/Admin users to configure security policies: MFA enforcement toggle, session timeout duration, and password policy.
REQ-SET-004	The system SHALL allow Technical/Admin users to configure compliance deadline reminder lead times (e.g. remind 14 days before, 7 days before, 1 day before).
REQ-SET-005	The system SHALL allow Technical/Admin users to export the full audit log for a selected date range in CSV format.

6. Business Workflows

6.1 Manual Data Submission Workflow

Step	Actor	Action	System Response
1	Participant Institution	Sends data file to RSwitch via agreed channel (email / secure transfer)	—
2	Compliance / Technical User	Logs in to RDIP and navigates to Compliance → Upload Submission	Upload form displayed with participant and requirement fields
3	Compliance / Technical User	Selects participant, requirement type, and uploads the received file	System validates file against expected schema
4	System	Schema validation passes	Submission recorded as "Submitted"; pipeline queued
4a	System	Schema validation fails	Error report returned; submission status = "Validation Failed"; Compliance user notified
5	Pipeline	Processes submitted file	Records ingested; counts and errors logged
6	Compliance User	Reviews submission	Status updated to "Under Review"
7	Compliance User	Approves or rejects	Status updated to "Approved" or "Rejected"

6.2 Incident Management Workflow

Step	Actor	Action	System Response
1	Compliance User	Creates incident record for a participant	Incident logged as "Open"; assigned owner notified
2	Assigned Owner	Acknowledges incident	Status → "Under Investigation"
3	Compliance User	Adds remediation action items	Action items created with due dates
4	Assigned Owner	Completes remediation	Marks action items as "Completed"
5	Compliance User	Reviews and closes	Status → "Resolved" then "Closed"
5a	System	Remediation due date passes without completion	Escalation notification sent to senior Compliance user

6.3 Governance Meeting Workflow

Step	Actor	Action	System Response
1	Compliance User	Creates meeting record	Meeting added to calendar; invites notified
2	Compliance User	Uploads minutes after meeting	Document stored; linked to meeting record
3	Compliance User	Records decisions and action items	Action items created with owners and due dates
4	System	7 days before due date	Reminder notification sent to action item owner
5	System	Due date passes without completion	Status → "Overdue"; escalation sent to committee chair
6	Owner	Marks action item complete	Status → "Completed"

6.4 Data Quality — Correction and Resubmission Workflow

Step	Actor	Action	System Response
1	Compliance / Technical User	Uploads file on behalf of participant	System runs schema and quality validation
2	System	Validation fails (missing fields, invalid format, inconsistent data)	Submission status → "Validation Failed"; structured error report generated listing each issue by row/field
3	Compliance User	Reviews error report	Contacts institution offline to request corrected data
4	Institution	Provides corrected file via agreed channel	—
5	Compliance / Technical User	Re-uploads corrected file	System re-validates against schema
6	System	Validation passes	Submission status → "Submitted"; original failed attempt retained in history for audit
6a	System	Validation fails again	New error report generated; process repeats from Step 3
7	Compliance User	Reviews and approves	Status → "Approved"; correction history preserved

> ***Data quality rule:*** *The system SHALL never partially ingest a submission. A file either passes full validation and is ingested completely, or it is rejected entirely. Partial ingestion is not permitted.*

6.5 Compliance Deadline Workflow

Step	Actor	Action	System Response
1	System	Compliance deadline configured	Entry appears in compliance calendar
2	System	Lead time reached (e.g. 14 days before)	Reminder notification sent to Compliance user
3	Compliance / Technical User	Receives data from institution and uploads on their behalf	Submission status → "Submitted"
4	System	Deadline passes without submission	Status → "Overdue"; escalation notification sent to senior Compliance user
5	Compliance User	Reviews and approves submission	Status → "Approved"

7. Module Breakdown

Module	Purpose	Primary Users
Authentication	Login, MFA, session management	All
Dashboard	Role-aware operational overview	All
Integrations	Catalog and manage participant data connections	Technical, Compliance
Pipeline Monitor	Real-time ingestion job tracking	Technical, Compliance
Standards & Mapping	Define field translation rules between formats	Technical
Participant 360	Unified institutional profile	Compliance, Business, Technical
Governance	Document repository, meeting management, action tracking	Compliance, Business
Compliance Monitoring	Deadline tracking, incidents, remediation	Compliance, Participants
AI Decision Support	Pattern detection, prioritization, trend summaries	Business, Compliance
Financial Inclusion	Inclusion metrics by institution type and region	Business
Reports	Generate, schedule, and export structured reports	Compliance, Business
Notifications	In-app and email event alerts	All
Users & Security	User accounts, roles, and audit log	Technical
Settings	System configuration and security policies	Technical

8. Feature Details

8.1 Participant 360 — Profile View

Attribute	Detail
What is it?	A single-screen view of everything RSwitch knows about one participant institution
Who uses it	Compliance users (view), Business users (view), Technical users (view)
Inputs	Participant ID or name (search)
Outputs	Consolidated profile: integration status, submission history, compliance status, incidents, technical performance, outstanding actions, risk indicator
Validation	Only RSwitch internal users can access participant profiles
Exceptions	If a participant has no submissions yet, display "No data submitted yet" with expected due date

8.2 Manual Data Upload

Attribute	Detail
What is it?	Allows RSwitch staff to upload data files received from participant institutions
Who uses it	Compliance User, Technical / Admin User
Inputs	Requirement type (dropdown), reporting period, file (CSV / Excel / XML, max 50 MB)
Outputs	Submission record created; file stored; validation report displayed
Validation	File must match expected schema; file size ≤ 50 MB; file type must be CSV, XLSX, or XML
Exceptions	If validation fails, return a line-by-line error report; do not ingest partial data

8.3 AI Alert — Unusual Pattern

Attribute	Detail
What is it?	An AI-generated flag indicating that a participant's metrics deviate from expected patterns
Who uses it	Business users, Compliance users
Inputs	Historical submission data, failure rates, incident history
Outputs	Alert card with: participant name, triggered indicator, baseline value, current value, deviation %, recommended investigation area
Validation	Output must link to the source data driving the alert
Exceptions	If historical data is insufficient (< 3 months), alert should be suppressed and a "Insufficient history" label shown

8.4 Compliance Submission Review

Attribute	Detail
What is it?	Compliance users review and approve or reject a participant's submission
Who uses it	Compliance User
Inputs	Submission record, reviewer notes (optional), approval/rejection decision
Outputs	Submission status updated; participant notified; compliance history updated
Validation	Rejection requires a non-empty reason note
Exceptions	If a submission is rejected, the compliance deadline status remains "Overdue" until a new submission is approved

9. Status Definitions

9.1 Integration / Pipeline Status

Status	Meaning
Healthy	Last transfer succeeded with no errors
Warning	Transfer succeeded but with validation warnings
Failed	Last transfer failed or was rejected
Running	Transfer currently in progress
Queued	Transfer scheduled but not yet started
Paused	Transfer manually paused by admin
Not Configured	No integration set up for this data source yet

9.2 Compliance Submission Status

Status	Meaning
Pending	Deadline not yet reached; submission not yet made
Submitted	File uploaded; awaiting review
Validation Failed	Uploaded file did not pass schema or data quality checks; error report generated
Pending Correction	Submission rejected or failed validation; corrected resubmission required
Under Review	Compliance user is reviewing a valid submission
Approved	Submission accepted
Rejected	Submission rejected by Compliance user after review; resubmission required
Overdue	Deadline passed without an approved submission

9.3 AI Alert Status

Status	Meaning
Generated	Alert created by the system; not yet seen by a user
Acknowledged	User has read the alert and logged an awareness note — required before any further action
Under Investigation	Assigned to a specific user who is actively investigating — cannot be reached without passing through Acknowledged
Resolved	Investigation complete; outcome documented
Escalated	Alert sat at Acknowledged beyond the configured threshold (default 3 days) without progress, or severity increased; senior user notified
Closed	Formally closed with a mandatory written resolution summary; read-only — cannot be reopened

Lifecycle rules:

- An alert MUST pass through Acknowledged before moving to Under Investigation — skipping is not permitted
- Auto-escalation triggers if an alert remains at Acknowledged for more than the configured number of days without progressing (default: 3 days)
- Closure always requires a resolution summary of at least 20 characters
- A closed alert cannot be reopened — if the issue recurs, a new alert is generated

9.4 Incident Status

Status	Meaning
Open	Incident raised; not yet assigned
Under Investigation	Assigned owner acknowledged
Remediation In Progress	Remediation actions are active
Resolved	Remediation complete; pending closure
Closed	Incident formally closed by Compliance user

9.4 Governance Action Item Status

Status	Meaning
Open	Action item created; not started

In Progress	Owner is working on it
Completed	Owner marked done
Overdue	Due date passed without completion

10. Data Requirements

> ***Flexibility note (per stakeholder feedback):*** *All data fields, entities, and configurations defined in this section are indicative. The actual fields required will be confirmed and refined during the first phase of working with participating institutions. The data model SHALL be designed to accommodate new fields and entities without requiring architectural changes. Fields marked as required may become optional and vice versa as real data availability is understood.*

10.1 Participant Institution

Field	Type	Notes
`participant_id`	UUID	System-generated
`name`	String	Full legal name
`short_name`	String	Display abbreviation
`institution_type`	Enum	Commercial Bank / E-money Issuer / MFI / SACCO
`country`	String	
`region`	String	
`registration_date`	Date	Date joined RSwitch network
`primary_contact_name`	String	
`primary_contact_email`	String	
`status`	Enum	Active / Inactive / Suspended
`risk_score`	Float	Calculated composite score 0–100

10.2 Integration Configuration

Field	Type	Notes
`integration_id`	UUID	
`participant_id`	FK → Participant	
`source_name`	String	e.g. "Core Banking", "Mobile Money"
`integration_mode`	Enum	Manual / Automated API
`endpoint_url`	String	Encrypted at rest (API mode only)
`credentials`	Encrypted	Stored encrypted; never displayed in plain text
`schedule_cron`	String	e.g. "0 0 * * *" (API mode)
`status`	Enum	Healthy / Warning / Failed / Not Configured
`last_sync_at`	DateTime	
`last_sync_records`	Integer	
`last_sync_errors`	Integer	

10.3 Compliance Requirement

Field	Type	Notes
`requirement_id`	UUID	
`participant_id`	FK → Participant	
`requirement_name`	String	
`regulatory_reference`	String	e.g. "BNR Directive 07/2024"
`frequency`	Enum	Monthly / Quarterly / Annual / Ad hoc
`due_date`	Date	
`submission_status`	Enum	See Section 9.2
`submitted_at`	DateTime	Nullable
`reviewed_by`	FK → User	Nullable
`approved_at`	DateTime	Nullable
`rejection_reason`	Text	Nullable

10.4 Incident

Field	Type	Notes
incident_id	UUID	
participant_id	FK → Participant	
raised_by	FK → User	
raised_at	DateTime	
description	Text	
severity	Enum	Low / Medium / High / Critical
status	Enum	See Section 9.3
assigned_owner	FK → User	
resolved_at	DateTime	Nullable
closed_at	DateTime	Nullable

10.5 User

Field	Type	Notes
user_id	UUID	
full_name	String	
email	String	Unique
role	Enum	compliance / business / technical / participant
participant_id	—	Not applicable; all users are RSwitch internal staff
status	Enum	Active / Inactive / Invited
mfa_enabled	Boolean	
last_login_at	DateTime	
created_at	DateTime	

10.6 Governance Document

Field	Type	Notes
document_id	UUID	
title	String	
category	Enum	Policy / Minutes / Decision / Working Paper / Other
committee	String	e.g. "Board", "Steering Committee"
version	String	e.g. "v2.0"
uploaded_by	FK → User	
uploaded_at	DateTime	
file_url	String	Stored internally; access controlled
access_level	Enum	Internal / Restricted

11. Validation Rules

11.1 User Password

Rule	Requirement
Minimum length	12 characters
Uppercase	At least 1 uppercase letter
Lowercase	At least 1 lowercase letter
Number	At least 1 digit
Special character	At least 1 special character (!@#$%^&*)
Not reused	Cannot match last 5 passwords

11.2 Email Address

Rule	Requirement
Format	Must match RFC 5322 email format
Uniqueness	Must be unique across all user accounts
Domain	No validation on domain (any work email accepted)

11.3 File Upload (Compliance Submission)

Rule	Requirement
Allowed types	CSV, XLSX, XML
Maximum size	50 MB
Schema validation	File columns must match the expected schema for the selected requirement type
Encoding	UTF-8 required for CSV files

11.4 Compliance Deadline

Rule	Requirement
Due date	Must be a future date when creating a new requirement
Frequency	If recurring, next due date auto-calculated after approval of current cycle
Rejection reason	Mandatory text field (minimum 20 characters) when rejecting a submission

11.5 Participant Risk Score

Rule	Requirement
Range	0 (lowest risk) to 100 (highest risk)
Components	Weighted average of: integration reliability (30%), compliance adherence rate (30%), incident frequency (20%), overdue submission rate (20%)
Minimum data	Score only displayed when at least 3 months of history is available
Recalculation	Score recalculated nightly

12. User Permissions Matrix

12.1 Module-Level Permissions

Module	Compliance	Business	Technical
Dashboard — View	✓	✓	✓
Integrations — View	✓	✓	✓
Integrations — Manage	✗	✗	✓
Integrations — Upload on behalf	✓	✗	✓
Pipeline Monitor — View	✓	✓	✓
Pipeline Monitor — Manage	✗	✗	✓
Standards & Mapping — View	✓	✗	✓
Standards & Mapping — Manage	✗	✗	✓
Participant 360 — View	✓	✓	✓
Governance — View	✓	✓	✓
Governance — Manage	✓	✗	✗
Compliance — View	✓	✓	✓
Compliance — Manage	✓	✗	✗
Compliance — Upload Submission	✓	✗	✓
AI Decision Support — View	✓	✓	✓
AI Decision Support — Manage	✗	✓	✗
Financial Inclusion — View	✓	✓	✓
Financial Inclusion — Manage	✗	✓	✗
Reports — View	✓	✓	✓
Reports — Generate	✓	✓	✗
Reports — Schedule	✓	✓	✗
Users & Security — Manage	✗	✗	✓
Settings — View	✗	✓	✓
Settings — Manage	✗	✗	✓
Audit Log — View	✓	✗	✓

13. Notifications

Trigger	Recipient	Method
Integration fails	Technical/Admin team	Email + In-app
Participant submission overdue	Compliance user	Email + In-app
Compliance submission rejected	Compliance user (to follow up with institution offline)	In-app
Compliance submission approved	Compliance user	In-app
Incident created	Assigned owner	Email + In-app
Incident remediation overdue	Senior Compliance user	Email + In-app
Governance action item due in 7 days	Action item owner	Email + In-app
Governance action item due in 1 day	Action item owner	Email + In-app
Governance action item overdue	Committee chair + Action item owner	Email + In-app
New user invited	Invited user	Email (with registration link)
User role changed	Affected user	Email + In-app
AI alert generated	Business users + Compliance users	In-app
Scheduled report generated	Requesting user	Email (with attachment)
Password reset requested	Requesting user	Email (with reset link)

14. Reports

14.1 Compliance Summary Report

Attribute	Detail
Purpose	Overview of compliance status across all or selected participants
Filters	Participant, institution type, date range, status (Pending / Overdue / Approved / Rejected)
Columns	Participant name, requirement name, due date, submission date, status, reviewer, rejection reason
Export formats	PDF, Excel
Permissions	Compliance (generate), Business (view)

14.2 Integration Health Report

Attribute	Detail
Purpose	Summary of all integration statuses and recent pipeline runs
Filters	Participant, integration type, date range, status
Columns	Participant, source, status, last sync date, records transferred, error count, uptime %
Export formats	PDF, Excel
Permissions	Compliance (view), Business (view), Technical (generate)

14.3 Participant Performance Report

Attribute	Detail
Purpose	Ranked view of all participants by operational performance
Filters	Institution type, region, date range
Columns	Participant, risk score, compliance rate, failure rate, open incidents, overdue submissions
Export formats	PDF, Excel
Permissions	Compliance (generate), Business (generate)

14.4 Financial Inclusion Report

Attribute	Detail
Purpose	Aggregated inclusion metrics by institution type and region
Filters	Institution type, region, period
Columns	Institution type, region, metric name, value, data source, last updated
Export formats	PDF, Excel
Permissions	Business (generate), Compliance (view)

14.5 Incident Report

Attribute	Detail
Purpose	List of all incidents within a period
Filters	Participant, severity, status, date range
Columns	Incident ID, participant, severity, description, raised date, status, resolved date, resolution summary
Export formats	PDF, Excel
Permissions	Compliance (generate)

15. Dashboard Widgets

15.1 Compliance User Dashboard

Widget	Description
Overdue Submissions	Count of compliance submissions past due date
Upcoming Deadlines	List of deadlines in next 30 days
Open Incidents	Count by severity (Critical / High / Medium / Low)
Compliance Rate (30d)	% of submissions approved on time across all participants
Overdue Action Items	Count of governance and incident action items past due
Recent Activity	Last 10 compliance events

15.2 Business User Dashboard

Widget	Description
Integration Health Summary	Count of Healthy / Warning / Failed integrations
AI Alerts (Unacknowledged)	Count of pending AI alerts
Financial Inclusion Snapshot	Key inclusion metrics for current period
Participant Risk Overview	Distribution of participant risk scores
Recent Reports	Last 5 generated reports with download links
Compliance Overview	Aggregate compliance rate across all participants

15.3 Technical / Admin User Dashboard

Widget	Description
Active Pipelines	Count running + count failed today
Failed Integrations	List of integrations currently in Failed state
Data Freshness	Last sync time per data source
System Health	Uptime, error rate, queue depth
Pending Mappings	Count of mappings in Draft status
Recent Pipeline Runs	Last 10 pipeline jobs with status

16. External Integrations

> ***Email scope clarification (per stakeholder feedback):*** *Email integration is outbound only. RDIP sends automated notifications and scheduled reports via email. The system does NOT ingest incoming emails, parse email replies, or process calendar invite acceptances. Any response from participants to email notifications is handled offline by RSwitch staff.*

System	Purpose	Direction	Authentication
Participant Core Banking System	Pull transaction summary data (automated mode)	Inbound to RDIP	API Key / OAuth 2.0
Participant Mobile Money Platform	Pull mobile money submission data (automated mode)	Inbound to RDIP	API Key
Email Service (SMTP / SendGrid)	Deliver automated notification and report emails — outbound only; no email ingestion	Outbound from RDIP only	SMTP credentials / API key
File Storage (Internal)	Store uploaded documents and generated reports	Internal	IAM role-based
AI/ML Analytical Engine	Process operational data and generate alerts; model-agnostic — engine is replaceable without code changes	Internal	Internal service token
National Regulatory Reporting System	Submit required regulatory reports (future)	Outbound from RDIP	TBD by regulator

17. Non-Functional Requirements

17.1 Performance

Requirement	Target
Page load time	< 2 seconds for 95% of requests under normal load
API response time	< 500 ms for standard queries
File upload processing	< 60 seconds for files up to 50 MB
Dashboard data refresh	< 3 seconds
Concurrent users	System must support at least 100 concurrent users without degradation
Report generation	Standard reports generated in < 30 seconds

17.2 Security

Requirement	Detail
Transport encryption	All communications over HTTPS / TLS 1.3
Data encryption at rest	All stored data encrypted using AES-256
Authentication	JWT-based session tokens with expiry
MFA	TOTP-based MFA mandatory for all users
RBAC	Role-based access control enforced server-side on every API endpoint
API security	Rate limiting: 100 requests/minute per user; 1000/minute per IP
Secrets storage	API keys and credentials stored encrypted; never logged or displayed in plain text
Audit logging	All data-modifying actions logged with user, timestamp, and IP
OWASP protection	Protection against SQL injection, XSS, CSRF, and the OWASP Top 10
Penetration testing	System must pass a penetration test before go-live

17.3 Scalability

Requirement	Detail
Horizontal scaling	Application tier must support horizontal scaling
Database	Read replicas supported for reporting queries
Caching	Frequently accessed reference data cached with a max 5-minute TTL
Queue system	Asynchronous job queue for pipeline execution and report generation

17.4 Reliability

Requirement	Target
Uptime	99.5% monthly uptime (excluding scheduled maintenance)
Scheduled maintenance	Maximum 4 hours/month; communicated 48 hours in advance
Backup frequency	Full database backup daily; transaction logs backed up every hour
Recovery Point Objective (RPO)	Maximum 1 hour of data loss in worst-case failure
Recovery Time Objective (RTO)	System restored within 4 hours of major failure
Monitoring	Real-time system health monitoring with alerting to Technical team

17.5 Accessibility

Requirement	Detail
Standard	WCAG 2.1 Level AA compliance
Keyboard navigation	All core workflows navigable by keyboard
Color contrast	Minimum 4.5:1 contrast ratio for all text
Browser support	Chrome 110+, Firefox 110+, Edge 110+, Safari 16+
Responsive design	Functional on screen widths ≥ 1024px (desktop-first)

17.6 DevOps & Deployment

17.6.1 Environments

Environment	Purpose
Development	Used by developers for active feature work; connected to mock data
Staging	Mirror of production used for QA, UAT, and pre-release testing
Production	Live system used by RSwitch staff; requires change approval before deployment

All three environments SHALL be isolated from each other. Staging SHALL use anonymized/synthetic data only — never real participant data.

17.6.2 CI/CD Pipeline

Requirement	Detail
Source control	All code SHALL be managed in a version control system (e.g. Git) with a defined branching strategy
Automated builds	Every code push to a main branch SHALL trigger an automated build
Automated testing	Unit and integration tests SHALL run automatically on every build; a failing test SHALL block deployment
Deployment to staging	Successful builds on the main branch SHALL be automatically deployed to the staging environment
Deployment to production	Production deployments SHALL require a manual approval step from the Technical Lead
Rollback	Every production deployment SHALL support one-click rollback to the previous stable version
Deployment downtime	Deployments SHALL be designed to achieve zero downtime using rolling or blue-green deployment strategies

17.6.3 Infrastructure & Hosting

Requirement	Detail
Initial hosting	Platform SHALL be deployed on RSwitch's internal server infrastructure for v1
Web server	A production-grade web server (e.g. Nginx) SHALL be configured as a reverse proxy
Process management	Application processes SHALL be managed by a process manager (e.g. PM2, systemd) to auto-restart on failure
Job queue	Background jobs (pipeline execution, report generation, notifications) SHALL run via an asynchronous queue system separate from the web server process
Scheduler	Recurring pipeline and report jobs SHALL be managed by a cron-based scheduler
SSL / TLS	All environments SHALL use HTTPS with valid TLS certificates; certificates SHALL be auto-renewed before expiry
Firewall	Production server SHALL allow only ports 80, 443 (web) and a restricted SSH port; all other ports SHALL be closed
Secrets management	All secrets (database passwords, API keys, JWT secrets) SHALL be stored in environment variables or a secrets manager — never hardcoded in source code

17.6.4 Monitoring & Logging

Requirement	Detail
Uptime monitoring	An external uptime monitor SHALL check the platform every 60 seconds and alert the Technical team if the platform is unreachable for more than 2 minutes
Error tracking	Application errors SHALL be captured and reported to an error tracking tool (e.g. Sentry) in real time
Performance monitoring	Server CPU, memory, and disk usage SHALL be monitored continuously; alerts triggered when CPU > 85% or disk > 80%
Pipeline failure alerts	Failed pipeline jobs SHALL trigger an immediate in-app and email notification to the Technical team
Centralized logging	All application logs (requests, errors, background jobs) SHALL be written to a centralized log store with a minimum 90-day retention
Audit log	All user actions that modify data SHALL be written to a tamper-evident audit log (see REQ-SET-005)
Log access	Logs SHALL be accessible to Technical/Admin users through the Settings module or directly on the server

17.6.5 Backup & Recovery

Requirement	Detail
Database backups	Full database backup SHALL run automatically every 24 hours
Transaction log backups	Database transaction logs SHALL be backed up every hour to support point-in-time recovery
Backup storage	Backups SHALL be stored in a separate location from the primary server (offsite or separate storage volume)
Backup retention	Daily backups SHALL be retained for 30 days; weekly backups for 6 months
Backup testing	Backup restore SHALL be tested at least once per quarter; results documented
Disaster recovery	A documented disaster recovery runbook SHALL exist before go-live, covering: server failure, data corruption, and full environment loss

RPO	Recovery Point Objective: maximum 1 hour of data loss
RTO	Recovery Time Objective: system restored within 4 hours of a major failure

18. Database Models

Key Entities

```
Participant
  ├── Integration[]
  │     └── PipelineJob[]
  ├── ComplianceRequirement[]
  │     └── Submission[]
  ├── Incident[]
  │     └── RemediationAction[]
  ├── ParticipantRiskScore (calculated)

GovernanceCommittee
  ├── Member[]
  ├── Meeting[]
  │     ├── Document[]
  │     └── ActionItem[]
  └── Document[]

FieldMapping
  ├── MappingVersion[]
  └── FieldRule[]

User
  ├── AuditLog[]
  └── Notification[]

Report
  └── ReportSchedule[]
```

19. API Requirements

19.1 Authentication

Method	Endpoint	Description
POST	/api/auth/login	Authenticate user, return JWT
POST	/api/auth/mfa/verify	Verify TOTP code, complete login
POST	/api/auth/logout	Invalidate session token
POST	/api/auth/password/reset-request	Send password reset email
POST	/api/auth/password/reset	Complete password reset with token

19.2 Participants

Method	Endpoint	Description
GET	/api/participants	List all participants (with filters)
GET	/api/participants/:id	Get participant profile (Participant 360)
POST	/api/participants	Create new participant
PUT	/api/participants/:id	Update participant details
GET	/api/participants/:id/submissions	List submission history
GET	/api/participants/:id/incidents	List incidents
GET	/api/participants/:id/risk-score	Get current risk score with breakdown

19.3 Compliance

Method	Endpoint	Description
GET	/api/compliance/requirements	List all compliance requirements
POST	/api/compliance/requirements	Create new requirement
POST	/api/compliance/submissions	Upload new submission
PUT	/api/compliance/submissions/:id/review	Approve or reject submission
GET	/api/compliance/incidents	List all incidents
POST	/api/compliance/incidents	Create new incident
PUT	/api/compliance/incidents/:id	Update incident status

19.4 Integrations & Pipelines

Method	Endpoint	Description
GET	/api/integrations	List all integrations
POST	/api/integrations	Create new integration
PUT	/api/integrations/:id	Update integration config
POST	/api/integrations/:id/sync	Trigger manual sync
GET	/api/pipelines/jobs	List pipeline jobs (with filters)
POST	/api/pipelines/jobs/:id/reprocess	Reprocess failed job
POST	/api/pipelines/jobs/:id/stop	Stop running job
GET	/api/pipelines/jobs/:id/logs	Get execution log

19.5 Governance

Method	Endpoint	Description
GET	/api/governance/documents	List governance documents
POST	/api/governance/documents	Upload new document
GET	/api/governance/meetings	List meetings
POST	/api/governance/meetings	Create meeting record
PUT	/api/governance/meetings/:id	Update meeting (add minutes, decisions)
GET	/api/governance/action-items	List action items
PUT	/api/governance/action-items/:id	Update action item status

19.6 Reports

Method	Endpoint	Description
GET	/api/reports	List generated reports
POST	/api/reports/generate	Generate new report
GET	/api/reports/:id/download	Download report file
POST	/api/reports/schedules	Create recurring report schedule
DELETE	/api/reports/schedules/:id	Cancel recurring schedule

19.7 Users

Method	Endpoint	Description
GET	/api/users	List all users
POST	/api/users/invite	Send invitation to new user
PUT	/api/users/:id/role	Change user role
PUT	/api/users/:id/deactivate	Deactivate user account
GET	/api/audit-log	Retrieve audit log entries

20. Error Handling

20.1 HTTP Error Codes

Code	Status	User-Facing Message
400	Bad Request	"The request could not be understood. Please check your input and try again."
401	Unauthorized	"You must be logged in to access this page."
403	Forbidden	"You do not have permission to perform this action."
404	Not Found	"The requested record could not be found."
409	Conflict	"This record already exists or conflicts with an existing entry."
413	Payload Too Large	"The uploaded file exceeds the 50 MB maximum size."
422	Unprocessable Entity	"The file did not pass validation. See the error report below."
429	Too Many Requests	"Too many requests. Please wait a moment before trying again."
500	Internal Server Error	"Something went wrong on our end. The technical team has been notified."
503	Service Unavailable	"The service is temporarily unavailable. Please try again in a few minutes."

20.2 Validation Errors

- Validation errors SHALL be returned as a structured JSON array identifying the failing field and the specific rule violated.
- The UI SHALL display field-level inline error messages next to the relevant input, not just a top-level banner.
- File validation errors SHALL include the row number, column name, and description of the violation.

21. Release Plan

Release 1 — Core Foundation (MVP)

Area	Features
Authentication	Login, MFA, password reset, RBAC
Dashboard	Role-aware KPI dashboard with attention queue
Integrations	Participant catalog, manual upload, integration status display
Pipeline Monitor	Job list, logs, status, reprocess
Participant 360	Basic profile with integration status and submission history
Compliance Monitoring	Compliance register, deadline tracking, submission upload and review
Users & Security	User invitation, role management, audit log
Notifications	Email + in-app for compliance deadlines and integration failures

Release 2 — Governance & Intelligence

Area	Features
Governance	Document repository, meeting calendar, action items
Standards & Mapping	Field mapping management, versioning
AI Decision Support	Unusual pattern alerts, participant priority list
Reports	All report types, PDF/Excel export, scheduling
Participant 360	Full profile including risk score, incidents, technical performance
Notifications	Full notification matrix including governance reminders

Release 3 — Advanced Analytics & Automation

Area	Features
Financial Inclusion	Dashboards segmented by institution type and region
Automated API Integrations	Scheduled pull from participant endpoints
AI Trend Summaries	Compliance and operational trend reports
Settings	Security policy configuration, API key management
Regulatory Reporting	Export to national regulatory systems (where defined)

22. Testing Requirements

Test Type	Scope	Acceptance Criteria
Unit Testing	All service-layer functions and validation rules	≥ 80% code coverage
Integration Testing	API endpoints, database operations, file processing	All endpoints return expected responses for valid and invalid inputs
API Testing	Every endpoint in Section 19	All request/response contracts validated; error codes correct
UI Testing	All screens across all 3 roles	Core workflows pass without errors in Chrome, Firefox, Edge
Security Testing	Authentication, RBAC, file upload, API rate limiting	Penetration test passes; no Critical or High findings unresolved
Load Testing	100 concurrent users performing dashboard load and report generation	Response time < 2 seconds at 95th percentile
User Acceptance Testing (UAT)	End-to-end workflows with RSwitch Compliance and Technical teams	All agreed UAT scenarios signed off by stakeholder
Data Validation Testing	File upload validation rules	All invalid file formats and schema violations rejected correctly
Infrastructure Testing	CI/CD pipeline, environment parity (staging vs production), SSL, firewall rules	Deployment pipeline runs end-to-end without manual steps; staging mirrors production config
Backup & Recovery Testing	Database restore from backup in staging environment	Full restore completes within RTO; restored data matches RPO window
Smoke Testing	Core user journeys (login, upload, report generation) run automatically after every production deployment	All smoke tests pass within 5 minutes of deployment; automatic rollback triggered on failure

23. Go-Live Checklist

Item	Responsible	Status
All Release 1 features complete and tested	Development Team	—
UAT sign-off received from RSwitch	Stakeholders	—
Security penetration test passed	Security Team	—
All Critical and High findings resolved	Development Team	—
DevOps		
Production environment provisioned and hardened	DevOps / Technical Team	—
Staging environment verified to match production config	DevOps / Technical Team	—
CI/CD pipeline fully operational (build → test → deploy)	DevOps / Technical Team	—
SSL certificate installed, valid, and auto-renewal configured	DevOps / Technical Team	—
Firewall rules configured; only ports 80 and 443 open	DevOps / Technical Team	—
Secrets moved out of source code into environment variables	DevOps / Technical Team	—
Uptime monitoring configured and alerting verified	DevOps / Technical Team	—
Error tracking (e.g. Sentry) active in production	DevOps / Technical Team	—
Server resource monitoring (CPU, memory, disk) enabled	DevOps / Technical Team	—
Database backups configured, automated, and tested restore verified	DevOps / Technical Team	—
Disaster recovery runbook documented and reviewed	DevOps / Technical Team	—
Background job queue and scheduler running and verified	DevOps / Technical Team	—
Smoke tests passing after production deployment	DevOps / Technical Team	—
Rollback procedure tested and documented	DevOps / Technical Team	—
Operations		
User accounts created for all RSwitch staff	Admin	—
System documentation complete	Project Team	—
User training delivered to all user groups	Project Team	—
Rollback plan documented and tested	Technical Team	—
Sample/synthetic data replaced with real data	RSwitch Team	—
Go-live approval from project sponsor	Stakeholders	—

24. Appendices

24.1 Glossary

Term	Definition
RDIP	RSwitch Data Intelligence Platform — the system described in this document
RSwitch	Rwanda's national payment switch operator
Participant	A financial institution connected to the RSwitch network (bank, e-money issuer, MFI, SACCO)
ETL	Extract, Transform, Load — the three-step process of moving and normalizing data
Pipeline	An automated ETL job that ingests data from a participant source into RDIP
Mapping	A defined set of rules that translate source data fields to a canonical target format
MFI	Microfinance Institution
SACCO	Savings and Credit Cooperative Organization
TOTP	Time-based One-Time Password — the standard behind authenticator app MFA codes
PAR	Portfolio at Risk — a financial metric measuring the proportion of a loan portfolio at risk of default
ISO 20022	International standard for electronic data interchange between financial institutions
RBAC	Role-Based Access Control
JWT	JSON Web Token — a standard for secure session tokens
KPI	Key Performance Indicator
UAT	User Acceptance Testing
WCAG	Web Content Accessibility Guidelines

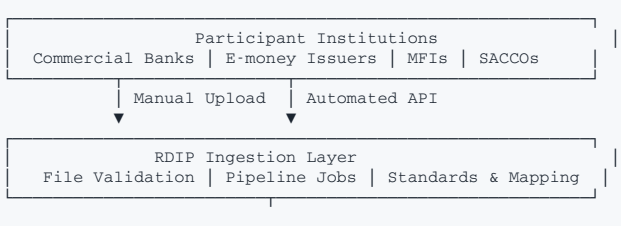
24.2 Acronyms

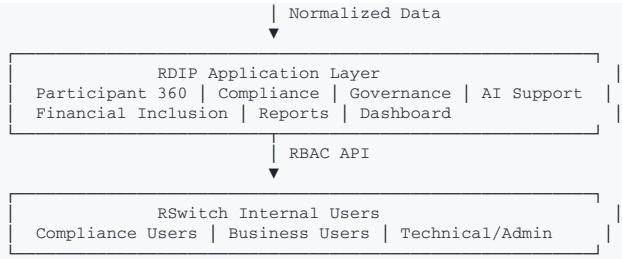
Acronym	Expansion
API	Application Programming Interface
CSV	Comma-Separated Values
XLSX	Microsoft Excel Open XML Spreadsheet
XML	Extensible Markup Language
PDF	Portable Document Format
HTTPS	Hypertext Transfer Protocol Secure
TLS	Transport Layer Security
AES	Advanced Encryption Standard
MFA	Multi-Factor Authentication
RBAC	Role-Based Access Control
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SRS	Software Requirements Specification
OWASP	Open Web Application Security Project

24.3 References

- RSwitch feedback document — RSwitch project review, July 2026
- ISO 20022 financial messaging standard — iso20022.org
- OWASP Top 10 Web Application Security Risks — owasp.org
- WCAG 2.1 Accessibility Guidelines — w3.org/WAI
- Rwanda Data Protection Law — RISA / NCSA

24.4 Architecture Diagram (Conceptual)





24.5 Key Open Items (Action Required)

Item	Owner	Notes
Confirm available data fields for Financial Inclusion dashboards	RSwitch Team	Required before finalizing FIN module metrics
Provide sample data for synthetic dataset generation	RSwitch Team	Required for development and testing
Confirm AI model or analytical approach	RSwitch Team + Project Team	Required before implementing AI Decision Support
Define regulatory reporting system API for Release 3	RSwitch + Regulator	Out of scope for v1; needs definition for Release 3
Confirm list of participant institutions for initial onboarding	RSwitch Compliance Team	Needed for UAT planning